📓 SOC Internship – Day 11 Notes

🛡️ MITRE ATT&CK Framework

🧠 1. What is MITRE ATT&CK?

MITRE ATT&CK is a **knowledge base of cyberattack techniques** based on real-world attacks.

👉 It helps SOC analysts:

- Understand attacker behavior
- Detect threats faster
- Respond effectively to incidents

🎯 2. Tactics (High-Level Goals of Attackers)

Tactics describe **what the attacker wants to achieve**.

◇ 1. Initial Access

- How attacker enters the system
- Example: Phishing email

◇ 2. Execution

- Running malicious code
- Example: Malware execution

◇ 3. Persistence

- Maintaining long-term access

- Example: Backdoor installation

◇ 4. Privilege Escalation

- Gaining higher (admin) access
- Example: Exploiting vulnerabilities

◇ 5. Credential Access

- Stealing usernames/passwords
- Example: Brute force attack

◇ 6. Lateral Movement

- Moving across systems in a network
- Example: Using stolen credentials

⚙️ 3. Techniques (How Attacks Are Done)

Techniques explain **how attackers perform actions**.

Examples:

- Brute force login
- Phishing emails
- Malicious script execution
- Credential dumping

🔎 4. Example Attack Mapping

Activity	MITRE Tactic	Technique
Multiple failed logins	Credential Access	Brute Force
Phishing email	Initial Access	Phishing

Malware execution	Execution	Malicious Code

5. Why MITRE ATT&CK is Important in SOC

- Helps understand attacker behavior
- Improves detection rules (SIEM alerts)
- Maps alerts to attack stages
- Strengthens incident response

Assignment Answers

1. Explain MITRE ATT&CK

MITRE ATT&CK is a framework that organizes real-world cyberattack methods into tactics and techniques, helping SOC analysts detect and respond to threats efficiently.

2. Three Tactics with Examples

- **Initial Access** → Phishing email
- **Execution** → Running malware
- **Credential Access** → Brute force login

3. Brute Force Attack Mapping

- **Tactic:** Credential Access
- **Technique:** Brute Force Login
- **Scenario:** Multiple failed login attempts to guess passwords